

UVC speed calculator — estimate, measure, then validate

1 Enter real geometry

- target plant dose
- active lamps / relay group
- plant distance
- UV output and aging
- exposure length
- measured utilization factor

2 Calculate estimate

$$I = N \times P \times \text{age} \times U \div (4\pi r^2)$$

$$\text{time} = \text{target dose} \div I$$

$$\text{speed} = \text{exposure length} \div \text{time}$$

Estimated speed and time

3 Measure and validate

Place UV-C radiometer at plant

Measure each group and position

Replace estimate with measured I

Run a controlled plant trial

Approve speed in site SOP